

Preface

Who is This Book For?

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Why Write Yet Another Book About Linear Algebra?

The direct answer to this question is simple: I really think most resources are doing a *bad job* of explaining fundamental linear algebra to new students ¹. Not because they lack material or the complete picture - but because the common approach to teaching fundamental mathematical topics just doesn't work well for most people ².

Introduction courses to fundamental mathematical topics usually look like this: here're some basic definitions, let's derive some specific results from them - now go apply these in your field.

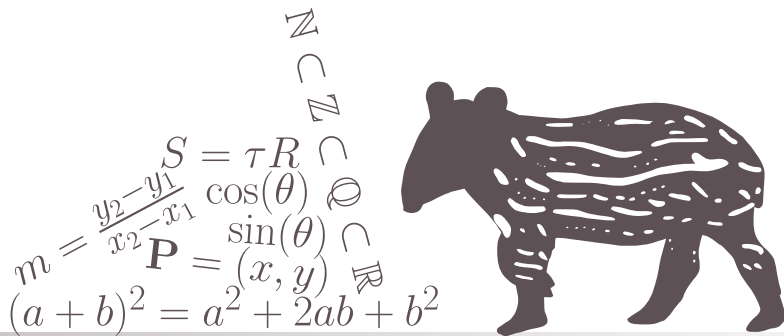
¹and honestly, this isn't restricted to linear algebra but applies to all fundamental mathematical topics - but it's a classic case

²the main exception is students of pure mathematics and people who can easily deal with extremely abstract concepts - a very specific group of people at the end of the day

CHAPTER

1

THE BASICS OF LINEAR ALGEBRA



1.1 VECTORS AND VECTOR SPACES

1.1.1 Motivation

Motivating Problem 1.1 Test

This is a test.



1.1.2 Geomteric Properties of Vectors

1.1.3 Geomteric Objects

1.1.4 Vector Spaces

1.1.5 Other Common Coordinate Systems in 2- and 3-Dimensions

1.1.6 Projections and Rejections

1.1.7 Solving the Motivating Problems

1.1. *VECTORS AND VECTOR SPACES*

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